



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

LIBRARY MANAGEMENT SYSTEM

The domain of the Project:

Salesforce Administration & Development

Team Mentors (and their designation):

Sriram Yarrabothula

Salesforce Developer

Submitted By:

Mr. ARAVALA VENKATESH

Period of the project

August 2026 to September 2026



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Declaration

The project titled “**Library Management System**” has been mentored by **Sriram Yarrabothula** and organized by **SURE Trust** from **April 2026 to September 2026**. The project was undertaken to provide educated unemployed rural youth with hands-on experience in working on **industry-relevant projects**, helping them gain practical knowledge and move closer to prospective employment opportunities. I declare that, to the best of my knowledge, the members of the team mentioned below have successfully worked on this project and enhanced their practical knowledge in the **Salesforce and Library Management domain**.

Submitted By:

Mr.ARAVALA VENKATESH

Mentor’s Name: Sriram Yarrabothula

Designation— Salesforce Developer Smartbridge

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

Project Overview — Library Management System

Objectives

The main objective of the **Library Management System** is to develop a centralized Salesforce-based platform for managing library operations and book-related activities. The project aims to:

- Maintain **book and library branch records**.
- Manage **member and librarian information**.
- Manage **book issue and return transactions**.
- Automate **book status updates, overdue tracking, and email notifications**.
- Manage **book reservations, suppliers, and fine payments**.
- Provide **role-based security** for Admin, Librarians, Assistant Librarians, and Members.
- Provide **dashboards and reports** for library management and decision-making.

Methods

The system was developed using **Salesforce declarative and programmatic features**, including:

- **Standard and Custom Objects, Fields, Relationships, and Validation Rules** for managing library data.
- **OWD, Role Hierarchy, and Profiles** for security and role-based access.
- **Screen Flows** for book issue and return processes.
- **Record-Triggered Flows** for automatic book status updates.
- **Scheduled Flows** for overdue book tracking and due-date reminder emails.
- **Apex Trigger** to prevent issuing unavailable books.
- **Batch Apex** for generating overdue book reports.
- **Schedulable Apex** for sending overdue notifications.
- **Lightning Web Components (LWC)** for the Book Dashboard, Issue Book/Return Book, and Reservation functionality.
- **Reports and Dashboards** for book availability, issued books, overdue books, book categories, and fine collection analysis.

Key Findings

The completed system demonstrates that Salesforce can effectively centralize library operations and reduce manual work. Key outcomes include:

- Book and library branch information can be maintained in a **structured and connected manner**.
- Members can be managed through **Contact records**.
- Book issue and return transactions can be tracked systematically.



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- Book status can be **automatically updated** when a book is issued or returned.
- The system can prevent users from issuing **unavailable books**.
- Overdue books can be identified automatically and **reminder notifications** can be generated.
- Members can reserve books and view their **reservation status** through the LWC.
- Fine information and payment status can be maintained for issued books.
- **Reports and dashboards** provide a consolidated view of library operations.
- **Role-based access** helps protect library and member information.



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Recommendations

For future enhancement, the Library Management System could include:

- Automated fine calculation and fine payment processing.
- More advanced book usage, borrowing, and overdue analytics.
- Mobile-friendly library functionality for members and librarians.
- Additional approval levels for book reservations and other library operations.
- Automated library alerts and reminders for due dates, overdue books, and reservations.
- Integration with external library systems, barcode scanners, RFID systems, or payment systems.
- More advanced analytics using Salesforce reporting and AI capabilities for predicting book demand and analyzing member borrowing patterns.

Conclusion

The project provides a centralized, secure, and automated Library Management System on Salesforce, integrating library branches, members, books, book issues, reservations, suppliers, fines, security, automation, Apex, Lightning Web Components, reports, and dashboards into a single platform.

Executive / Stakeholder Highlights

- **Centralized Library Management:** A single Salesforce platform manages library branches, books, members, book issues, reservations, suppliers, and fines, reducing fragmented library processes.
- **Strong Data Structure:** Related Salesforce objects create a connected library data model, linking library branches with books, members with book issues, books with issues and reservations, and issues with fines.
- **Improved Data Quality:** Validation rules help prevent incorrect data, including invalid due dates, invalid fine amounts, missing ISBN values, incorrect return dates, and duplicate active reservations.
- **Secure Access:** Private OWD, role hierarchy, and separate System Administrator, Librarian, and Member User profiles provide controlled access to library information.
- **Process Automation:** Salesforce Flows automate important library activities such as issuing and returning books, updating book status, identifying overdue books, and sending due-date reminder emails.
- **Book Availability Management:** Apex Trigger automation helps prevent users from issuing books that are unavailable, improving the accuracy of book issue transactions.
- **Member Self-Service:** Lightning Web Components allow members to view book availability, issue/return information, reserve books, and check reservation status.
- **Management Insights:** Reports and dashboards provide visibility into books by category, issued books, overdue books, book availability, and fine collection, supporting faster library management decisions.

Project Background, Goals, Scope & Innovation

1. Background and Context of the Project

The Library Management System is a Salesforce-based library management solution designed to centralize key library-related activities. The system manages library branches, books, members, book issues, reservations, suppliers, and fines within one platform. It also uses Salesforce automation, security features, Apex, Lightning Web Components, and reports/dashboards to support library operations.

2. Problem Statement / Goals

Traditional library processes can involve separate records and significant manual work for book management, member management, book issuing, returning, reservations, and fine tracking.



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The goal of this project is to provide a centralized and automated system that:

- **Maintains accurate library branch, book, and member records.**
- **Tracks book issue and return transactions.**
- **Manages and monitors book reservations.**
- **Automates book status updates.**
- **Identifies overdue books and sends reminder notifications.**
- **Prevents issuing of unavailable books.**
- **Manages fine information and collection.**
- **Provides secure, role-based access.**
- **Gives library management useful reports and dashboards.**

3. Scope and Limitations

Scope

- Library branch and book management.
- Member management.
- Book issue and return management.
- Book reservation management.
- Supplier and fine management.
- Role-based security.
- Salesforce Flows and Apex automation.
- Member self-service through LWCs.
- Reports and dashboards for library analytics.

Limitations

- The documented project does not specify integration with external library management, barcode, or RFID systems.
- Advanced AI-based library analytics are not part of the defined implementation.
- External payment gateway integration for fine payments is not part of the defined implementation.
- The system's functionality is primarily based on the Salesforce platform and its configured objects, automation, Apex, and Lightning Web Components.

4. Innovation Component

The innovation of the project is the integration of multiple library functions into one automated Salesforce ecosystem rather than managing each activity separately.

Key innovative aspects include:

- **Automated book issue and return management:** Screen Flows support the process of issuing and returning books.
- **Automated book status management:** Record-triggered automation updates the Book Status between Issued and Available based on book transactions.
- **Automated overdue management:** Scheduled automation identifies overdue books and supports due-date and overdue notifications.
- **Book availability control:** Apex Trigger logic prevents users from issuing a book when it is not available.
- **Member self-service:** LWCs provide members with direct access to book information, reservations, and reservation status.
- **Real-time management visibility:** Reports and dashboards provide consolidated insights into book availability, issued books, overdue books, categories, and fine collection.



Project Objectives

a) Clearly Defined Objectives and Goals of the Project

- **Primary Objective:** Build a comprehensive system to effectively manage **library branches, books, members, book issues, reservations, suppliers, and fines.**
 - **Data Architecture Goals:** Establish a robust data model by using **standard and custom Salesforce objects** and mapping their relationships, including **1-to-Many Lookup relationships** between Library Branches, Books, Members, Book Issues, Reservations, and Fines.
 - **Data Quality & Security Goals:** Enforce data integrity using **Validation Rules and Record Types**, while securing the system using a **Private Organization-Wide Default (OWD)**, **defined Role Hierarchy**, and **custom Profiles**.
 - **Automation Goals:** Streamline manual library processes using Salesforce **Flows (Screen, Record-Triggered, and Scheduled)** and programmatic **Apex (Trigger, Batch Apex, and Schedulable Apex)**.
 - **User Interface & Analytics Goals:** Develop custom user interfaces using **Lightning Web Components (LWC)** and visualize library data through dedicated **Reports and Dashboards**.
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b) Expected Outcomes and Deliverables

- **Database Deliverables:** A fully configured Salesforce environment containing the required **standard and custom objects**, including **Library Branch (Account)**, **Member (Contact)**, **Book**, **Book Issue**, **Reservation**, **Supplier**, and **Fine**, with their corresponding fields and relationships.
- **Declarative Automation Outcomes:** Active **Screen Flows** for issuing and returning books, **Record-Triggered Flows** to automatically update Book Status when books are issued or returned, and **Scheduled Flows** to identify overdue books and send due-date reminder notifications.
- **Programmatic (Apex) Outcomes:** Deployed Apex code including a **Trigger to prevent issuing unavailable books**, **Batch Apex for overdue book processing/reporting**, and **Schedulable Apex for overdue notifications**.
- **Custom UI Deliverables:** Three functioning **Lightning Web Components (LWC)**:
 - **Book Dashboard**
 - **Issue Book Component**
 - **Reservation Component**
- **Analytics Deliverables:** Dedicated reporting tools including **Books by Category**, **Issued Books**, **Overdue Books**, and **Fine Collection**, along with a centralized **Library Management Dashboard** displaying key metrics such as **Total Books**, **Available Books**, **Issued Books**, and **Total Fine Collection**.



Methodology and Results

Methodology and Technology Used

- **Salesforce Administration:** Utilized for core data modeling, creating **standard and custom objects, fields, object relationships, validation rules, record types, and security architecture.**
- **Declarative Automation:** Employed Salesforce's declarative, no-code methodology using **Screen Flows, Record-Triggered Flows, and Scheduled Flows** to automate book issue/return processes, book status updates, overdue tracking, and reminder notifications.
- **Programmatic Development:** Implemented **Apex programming** for complex backend logic, including a **Trigger to prevent issuing unavailable books, Batch Apex for overdue book processing, and Schedulable Apex for overdue notifications.**
- **Front-End Development:** Utilized the **Lightning Web Components (LWC)** framework to build dynamic and custom user interfaces for the **Book Dashboard, Issue Book/Return Book functionality, and Reservation Component.**

Tools/Software Used

- **Salesforce Lightning Platform:** The core cloud environment used to build and host the entire Library Management System.
- **Salesforce Flow Builder:** The tool used to construct **Screen Flows, Record-Triggered Flows, and Scheduled Flows** for library process automation.
- **Apex:** The object-oriented programming language used to implement the **Book Availability Trigger, Overdue Batch Apex, and Schedulable Apex.**
- **Lightning Web Components (LWC):** The UI framework utilized to build the **Book Dashboard, Issue Book Component, and Reservation Component.**
- **Salesforce Reports & Dashboards:** The native analytics tools used to generate visual summaries of **book categories, issued books, overdue books, book availability, and fine collection.**

Data Collection Approach

- **Interactive User Interfaces:** Library and book-related information can be entered and managed by authorized users through Salesforce records and custom **LWC interfaces.**
- **Book Issue and Return Components:** Book issue and return information is managed through the custom **Issue Book Component and Screen Flows**, allowing users to record transactions systematically.
- **Self-Service Components:** Reservation information can be submitted and monitored through the custom **Reservation Lightning Web Component.**
- **Automated System Processing:** Book status and overdue-related data are systematically processed in the background through **Record-Triggered Flows, Scheduled Flows, Batch Apex, and Schedulable Apex**, reducing manual processing.

Project Architecture

The system architecture is a **multi-layered Salesforce application** structured to ensure data integrity, security, automated processing, and user-friendly access.

Data Layer

The foundation is built using **standard and custom Salesforce objects**, including:

- **Account — Library Branch**
- **Contact — Member**
- **Book__c**
- **Book_Issue__c**
- **Reservation__c**
- **Supplier__c**
- **Fine__c**

The relational architecture uses **Lookup relationships** to connect library branches with books, members



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with book issues, books with book issues and reservations, and book issues with fines. Data quality is maintained through **Validation Rules**, while **Record Types** categorize Books as Physical/Digital and Members as Student/Faculty.

Security Layer

Access control is implemented using a **Private Organization-Wide Default (OWD)**. Visibility is managed through the defined **Role Hierarchy: Admin → Librarian → Assistant Librarian → Member**, while object-level access is controlled through the configured **System Administrator, Librarian, and Member User Profiles**.

Automation Layer

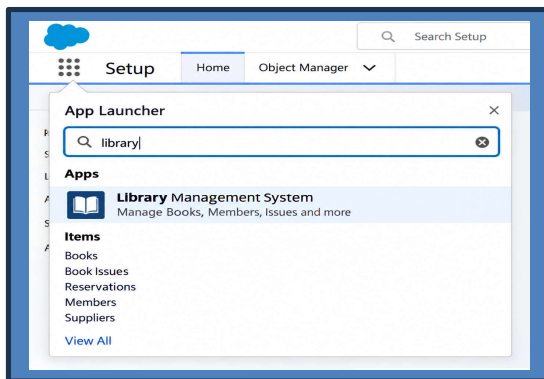
Business processes are handled using both **declarative automation and programmatic Apex**.

- **Flows** handle book issue and return processes, automatic Book Status updates, overdue identification, and reminder notifications.
- **Apex Trigger** prevents users from issuing unavailable books.
- **Batch Apex** supports overdue book processing/reporting.
- **Schedulable Apex** supports scheduled overdue notifications.

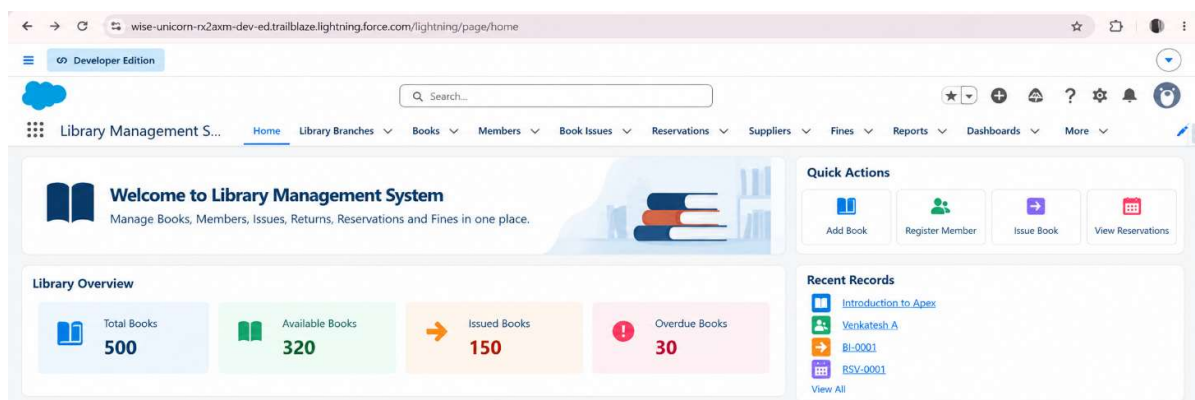
Presentation Layer

The presentation layer provides custom interfaces through **Lightning Web Components**, including the **Book Dashboard, Issue Book Component, and Reservation Component**. Salesforce **Reports and Dashboards** provide management-level visibility into book availability, issued books, overdue books, categories, and fine collection.

Final project working screenshots along with supporting explanation

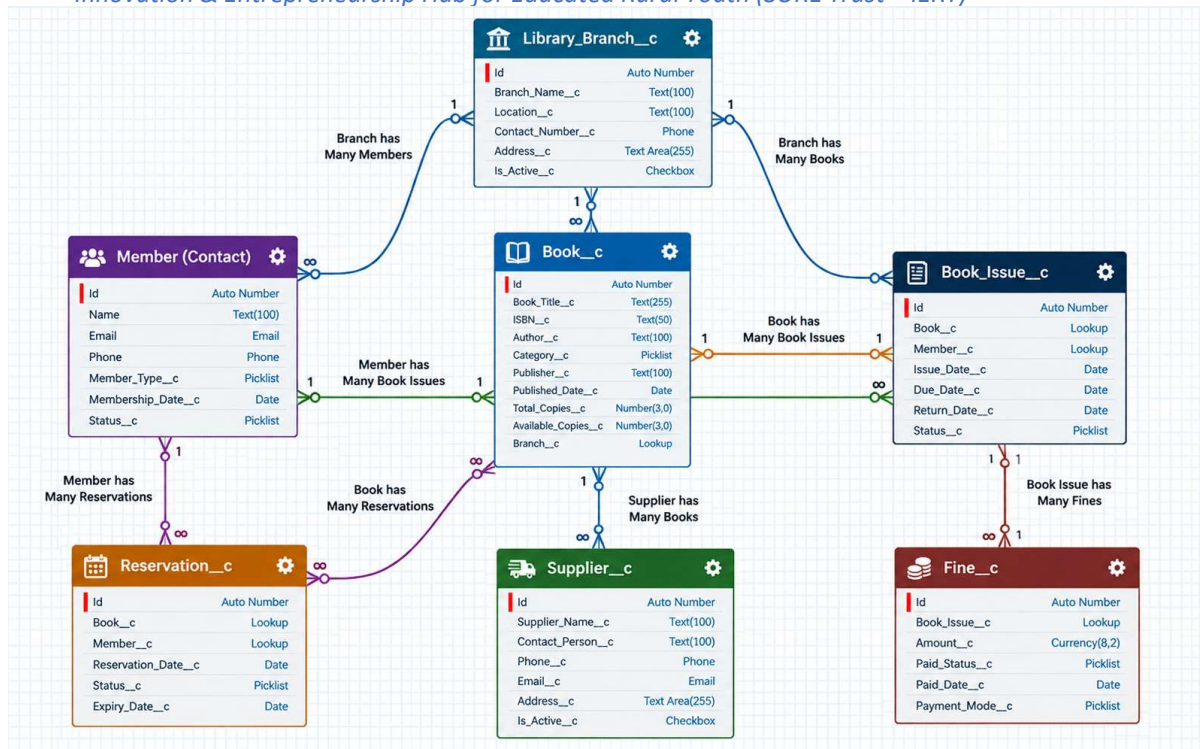


Title: Library Management System - App Home Page





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Title: Data Model (Salesforce Schema Builder)

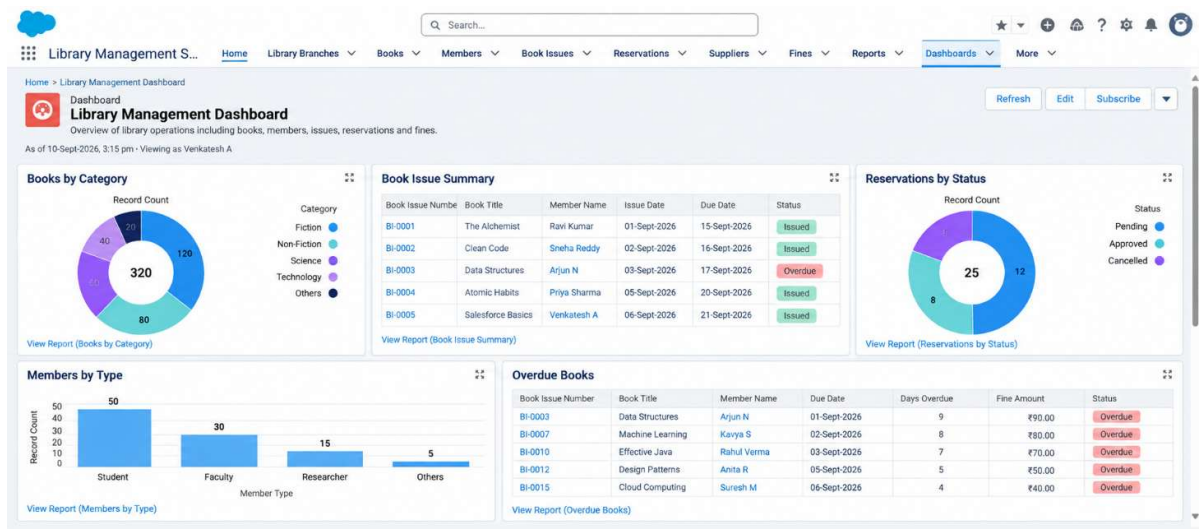
The screenshot shows the Salesforce Flow Builder interface. The top navigation bar includes the following tabs: Home, Library Branches, Books, Members, Book Issues, Reservations, Suppliers, Fines, Reports, Dashboards, Flows, and More. The "Flows" tab is selected, and the "Recently Viewed" section is displayed. The list contains 5 items, all of which are "Active" and were last modified by "Venkatesh A".

	Flow Label	Flow Type	Status	Last Modified Date	Last Modified By
1	Book Issue Flow	Screen Flow	Active	09/10/2026, 10:15 AM	Venkatesh A
2	Book Return Flow	Screen Flow	Active	09/09/2026, 4:32 PM	Venkatesh A
3	Update Book Status	Record-Triggered Flow	Active	09/08/2026, 3:21 PM	Venkatesh A
4	Overdue Books Notification	Scheduled Flow	Active	09/07/2026, 6:10 PM	Venkatesh A
5	Member Registration Flow	Screen Flow	Active	09/06/2026, 2:45 PM	Venkatesh A



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Title: Flow (Automation)



Title: Dash Board & Report

- Project GitHub Link



Learning and Reflection

Overall Experience

Working on the **Library Management System** under the mentorship of **Sriram Yarrabothula** has been a highly transformative professional journey. This project effectively bridged the gap between theoretical Salesforce concepts and hands-on application development by converting library business requirements into a functional, secure, and automated cloud application. One of the most rewarding aspects was troubleshooting and resolving configuration and automation issues during the implementation, which strengthened my debugging skills, problem-solving ability, and confidence in working with Salesforce. Ultimately, this experience strengthened my understanding of Salesforce development and prepared me for a professional role in the Salesforce domain.

New Learnings

From a technical perspective, I gained practical knowledge of **Salesforce Administration**, including creating standard and custom objects, configuring fields, establishing Lookup relationships, implementing Validation Rules, and using Record Types to organize library data. I gained hands-on experience with **Salesforce Flow Builder**, using Screen Flows for book issue and return processes, Record-Triggered Flows for automatic book status updates, and Scheduled Flows for overdue tracking and reminder notifications.

I also gained exposure to **Apex programming** by understanding and implementing an Apex Trigger to prevent issuing unavailable books, along with Batch Apex for overdue book processing and Schedulable Apex for scheduled overdue notifications. Furthermore, I learned how to develop custom user interfaces using **Lightning Web Components (LWC)**, including the Book Dashboard, Issue Book Component, and Reservation Component. I also strengthened my understanding of relational data modeling by connecting **Library Branches, Members, Books, Book Issues, Reservations, Suppliers, and Fines** through appropriate Salesforce relationships.

On the management side, I learned about **data security and governance**, particularly how to protect library information using a **Private Organization-Wide Default (OWD), Role Hierarchy, and custom Profiles**. I also developed project execution skills by dividing the project into manageable phases such as **Data Model, Security, Automation, Apex, User Interface, and Analytics**. Additionally, I learned to approach technical requirements from a business perspective, ensuring that validation rules, automation, and reports directly support efficient library operations.

Conclusion and Future Scope

Recap Objectives and Achievements

The primary objective of this project was to develop a **centralized, secure, and automated Library Management System on Salesforce** to streamline library operations. This objective was successfully addressed by creating a connected data model consisting of **Library Branch, Member, Book, Book Issue, Reservation, Supplier, and Fine** records.

Key achievements include the successful implementation of **Validation Rules and Record Types** for data integrity, along with a security model based on **Private OWD, Role Hierarchy, and Profiles**. The project also introduced automation through **Screen Flows for book issue and return, Record-**



Triggered Flows for book status updates, Scheduled Flows for overdue processing and reminders, and Apex components for book availability and overdue processing.

The development of **Lightning Web Components** further enhanced the user experience by providing custom interfaces for book dashboards, book issue/return operations, and reservations. Finally, **Reports and Dashboards** were implemented to provide insights into book categories, issued books, overdue books, reservations, and fine collection.

Future Scope of this Project

While the current system provides a strong foundation for managing library operations, there is substantial scope for future enhancement. Future development could focus on expanding the **Lightning Web Components (LWC)** to provide more advanced member self-service features, including detailed borrowing history, reservation tracking, and fine information.

The system could also be enhanced by integrating **barcode or RFID-based book tracking**, external payment systems for fine collection, and external library or communication platforms. Backend processing could be expanded to support more advanced **fine calculations, overdue processing, and large-scale library data management**. Additionally, Salesforce reporting and AI capabilities could be used to provide advanced insights such as **book demand analysis, borrowing trends, and recommendations for frequently requested books**.